Muhammed Ubada Alioğlu

Mechatronics Engineer · Automation Engineer · Konya, Türkiye
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Summary

Mechatronics engineer (Selçuk University, ranked first in department) working as an Automation Engineer at Arkon Otomasyon while pursuing a thesis-based MSc in Mechatronics Engineering. Writes PLC software on Siemens and CODESYS / Lenze platforms, designs HMI screens in TIA Portal, iX Developer, JMobile, WECON and Haiwell, and builds PC-based SCADA systems in C# / WPF over OPC UA — from step-sequence machine logic and servo motion to Modbus field devices and on-site troubleshooting.

Skills

PLC: Siemens S7-1200 (TIA Portal), Lenze PLC Designer / CODESYS (c300, c430), Beijer (BCS Tools)
PLC Languages: Structured Text (ST), Function Block Diagram (FBD), state-machine / step-sequence design
HMI & SCADA: TIA Portal, iX Developer, JMobile, WECON, Haiwell, custom SCADA in C# / WPF (.NET 8)
Communication: OPC UA, Modbus RTU / TCP, EtherCAT I/O, 4–20 mA / 0–10 V analog signals
Motion & Drives: Lenze i550 / i750 drives, servo axes (PLCopen motion), homing, PID control
Software: C#, Python
Design & Hardware: SolidWorks, STM32
Languages: Turkish (Advanced), Arabic (Advanced), English (Basic)

Experience

Arkon Otomasyon **Jan 2026 – Present**
Automation Engineer *Konya, Türkiye*

- Program and configure industrial systems on **Siemens** and **CODESYS / Lenze** platforms, writing machine logic in **Structured Text**.
- Design HMI screens in **TIA Portal**, **iX Developer**, **JMobile**, **WECON** and **Haiwell** that make machine control clearer for operators.
- Develop PC-based **SCADA** applications in **C# / WPF** communicating with PLCs over **OPC UA**, replacing legacy HMI / SCADA systems.
- Integrate field devices such as servo drives, energy analyzers, weighing transmitters and barcode readers over **Modbus** and analog I/O, and diagnose faults on site.

Arkon Otomasyon **Jun 2025 – Jul 2025**
Intern *Konya, Türkiye*

MES Mühendislik **Jul 2024 – Aug 2024**
Intern *Konya, Türkiye*

- Supported the technical team in **KNX**-based building automation and smart-home projects by observing on-site installation and commissioning.

Projects

Galvanizing Wash Line — SCADA Renewal | *Lenze PLC, C# / WPF, OPC UA* **2026**

- Built a new 1920×1080 **WPF SCADA** over **Lenze OPC UA** to replace the line's **iX Developer** HMI, keeping the original machine-picture layout operators rely on.
- Added a **product recipe log** (lot no, product, customer, raw material, acid / rinse / flux times), automatic furnace temperature and dip time, weekly pre-cleaning alarms and **PDF / Excel** reports.

Phosphating Machine — SCADA & Energy Analysis | *Lenze c300, OPC UA, Modbus, C# / WPF* **2026**

- Replaced an Exor HMI with a PC SCADA that calculates **energy consumption and cycle time per product** using a **Modbus** network analyzer.
- Implemented **barcode** product entry, three user levels (Admin / Manager / Operator) and a restricted **operator performance** page (output per hour / kg).

Wastewater Pump Stations — SCADA & PLC Control | *Beijer XP325, OPC UA, .NET 8 WPF* **2026**

- Replacing a Phoenix Visu+ SCADA for **4 pump stations** with a new **.NET 8 WPF / OPC UA** SCADA on the customer's offline network, with password-protected web access.
- Redesigned the PLC control: **3-stage pumping** driven by level with **PID** speed control (30–50 Hz), duty rotation, automatic standby on fault and **torque protection** with reverse-run cleaning.

Horizontal Flow-Pack Packaging Machine | *Lenze c430, 3 servo axes, Haiwell HMI* **2026**

- Programming a 3-servo machine (sealing jaw, pusher, film) on **Lenze c430**, including sensor-based **homing** sequences, 3 heating zones and a **Haiwell** HMI over **OPC UA**.

Eccentric Press Feeding Cell | *CODESYS ST, servo axis* **2026**

- Wrote the **step-sequence** logic for a servo axis, gripper, press and tray; separated the press cycle into an independent sequence so the axis returns while the press runs.
- Added a stop request that finishes the current part, live HMI status texts and adjustable timings.

Other Arkon Projects

- **Depositor machine** HMI (2 axes, position teach), unified **Lenze i550 / i750** error-handling function block, weighing indicator replacement with **4–20 mA** integration, pipe bending, feed pushing, bagging and Fellow machines.

University Projects **Selçuk University**

- **Teknofest Robolig** competition · **Drone** design and build · Design and mechanical calculation of an **oil pumping unit** · **Self-balancing bike**.

Education

Selçuk University, Konya — *MSc in Mechatronics Engineering (thesis, ongoing)*, **2026 – Present**
Selçuk University, Konya — *BSc in Mechatronics Engineering*, **2022 – 2026**

- Ranked **first in department** · High Honor student · GPA **3.54 / 4.00**

Certifications

PLC and Operator Panel Programming — Level 2 | *KTO Karatay University*, **2025**
PLC and Operator Panel Programming — Level 1 | *KTO Karatay University*, **2024**
Software Technologies and Artificial Intelligence | *GEN Academy*, **2025**